

2(a)(i)	specks of light moving haphazardly	B1
2(a)(ii)	(gas) molecules collide with (smoke) particles or random motion of the (gas) molecules	M1
	<u>causes</u> the (haphazard) motion of the smoke particles or <u>causes</u> the smoke particles to change direction	A1
2(b)(i)	$pV = nRT$	C1
	$n = (3.51 \times 10^5 \times 2.40 \times 10^{-3}) / (8.31 \times 290)$ or $n = (3.75 \times 10^5 \times 2.40 \times 10^{-3}) / (8.31 \times 310)$	C1
	or	
	$pV = NkT$	(C1)
	$n = (3.51 \times 10^5 \times 2.40 \times 10^{-3}) / (1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 290)$ or $n = (3.75 \times 10^5 \times 2.40 \times 10^{-3}) / (1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 310)$	(C1)
	$n = 0.350 \text{ mol}$ or 0.349 mol	A1
2(b)(ii)	energy transfer = $(0.349 \text{ or } 0.35) \times 12.5 \times (310 - 290)$	C1
	= 87.3 J or 87.5 J	A1
2(c)(i)	zero	A1
2(c)(ii)	87.3 J or 87.5 J	A1
	increase	B1